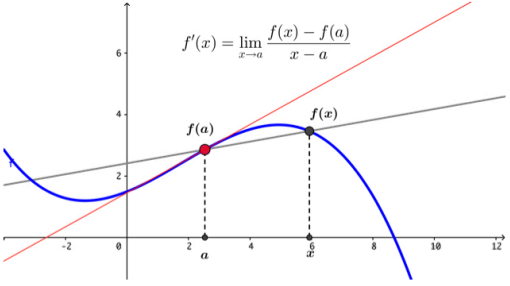
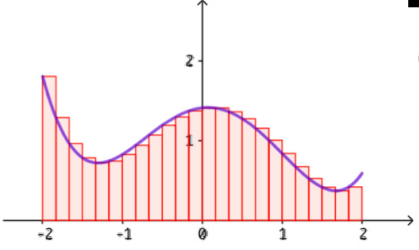
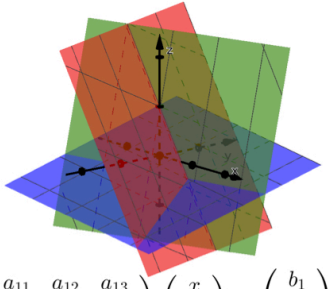


MATH1051

Calculus & Linear Algebra I



Week 8: Power Series (IN)



$$\int_a^b f(x)dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(c_i)\,\Delta x$$

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ b_2 \end{pmatrix}$$

Today's objectives

Part 1 (20 min)

- Introduction, Feedback & Summary
- Warm up questions

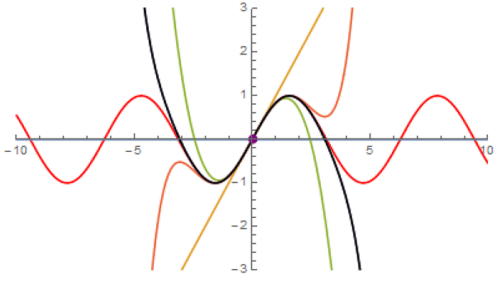
Part 2 (30 min)

- Problem Sheet 6 (Power Series)

Part 3 (55 min)

- Collaborative Exercise

$f(x) = \sin(x)$



$T_8(x) = x - \frac{x^3}{6} + \frac{x^5}{120} - \frac{x^7}{5040}$

Week 7 Exercise Feedback

- Marks for graded scripts are available on Blackboard > Gradebook.
- Report missing marks ASAP using a private post on the Ed Discussion Board
- Common errors/misconceptions.

Power Series Competency Checklist

PS1 I can determine the radius and interval of convergence of a simple power series.

PS2 I can recall standard Maclaurin series listed in the workbook and use them to construct Maclaurin series of simple functions.

PS3 I can construct the Taylor series of a simple function $f(x)$ about $x = a$ using the Taylor series formula. I recognise that the Maclaurin series is the Taylor series centered at $x = 0$.

A power series is a series of the form

$$f(x) = \sum_{n=0}^{\infty} c_n(x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + \dots$$

This is a *power series about* a. a and c_n's are fixed, x variable.

Such series have many applications, for example, solutions of higher order O.D.E's (ordinary differential equations) with non-constant coefficients, which in turn are very important to many areas of science and engineering.

(Stewart, p. 789) For a given power series $\sum_{n=0}^{\infty} c_n(x-a)^n$ there are only three possibilities:

1. The series converges only when $x = a$.
2. The series converges for all x.
3. There is a positive number r (called the *radius of convergence*) such that the series converges if $|x-a| < r$ and diverges if $|x-a| > r$.

Radius of convergence

7.12 The Ratio Test

(Stewart, p. 779) A powerful test for convergence of series is the ratio test. We will use it extensively in the following sections.

1. If $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = L < 1$, then the series $\sum_{n=1}^{\infty} a_n$ is absolutely convergent (and therefore convergent).
2. If $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = L > 1$ or $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \infty$, then the series $\sum_{n=1}^{\infty} a_n$ is divergent.
3. If $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = 1$, then the ratio test is inconclusive.
4. If $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$ is not defined, then the ratio test is inconclusive.

Taylor Series (Section 8.3)

If $f(x)$ has a power series representation at $x = a$, then

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n. \quad \text{for } |x-a|<r$$

$f^{(n)}(a)$ is the n-th derivative of $f(x)$ at $x = a$

$$\boxed{\textit{Taylor series of } f(x) \text{ about } x = a.}$$

Standard Maclaurin Series $a=0$

For $-1 < x < 1$,

$$\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k = 1 + x + x^2 + x^3 + x^4 + x^5 + \cdots.$$

For all x,

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \cdots.$$

For all x,

$$\sin x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots.$$

For all x,

$$\cos x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots.$$

For all $-1 < x \leq 1$,

$$\ln(1+x) = \sum_{k=1}^{\infty} (-1)^{k-1} \frac{x^k}{k} = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} - \cdots.$$

1. p-series test
2. t-series (Telescoping series)
3. g-series (Geometric series)
4. a-series (alternating series)
5. n-th term ***DIVERGENCE*** test
6. Comparison test
7. Ratio Test
8. Absolute convergence

Past Competency Questions

15 mins

1. Find the radius of convergence of the following power series: $\sum_{n=0}^{\infty} \frac{(3x-4)^n}{n!}$

2. What is the x^3 term in the Maclaurin series for $\sin x$?

3. The Taylor series of $f(x) = \frac{1}{1-x}$ at $x = 0$ is $\sum_{n=0}^{\infty} x^n$.

   What is the coefficient of x^3 in the Taylor series of $h(x) = \frac{1}{1-2x}$ at $x = 0$?

4. The MacLaurin series of $g(x) = \cos x$ is $\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n}$.

   What is the Maclaurin series of $h(x) = x^2 \cos x$?

1. Find the radius of convergence of the following power series: $\sum_{n=0}^{\infty} \frac{(3x-4)^n}{n!}$ $\quad a_n = \frac{(3x-4)^n}{n!}$

$\Rightarrow$ interval where the series is convergent $\Rightarrow$ use ratio test first

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\frac{\cancel{(3x-4)}^{n+1}}{\cancel{(n+1)!}}\cdot\frac{\cancel{n!}}{\cancel{(3x-4)^n}}\right| = \lim_{n\to\infty}\left|\frac{(3x-4)^1}{n+1}\right| = |3x-4|\cdot\lim_{n\to\infty}\left|\frac{1}{n+1}\right|$$

$$= |3x-4|\cdot 0 = 0 < 1$$

$\Rightarrow$ Power series is convergent $\forall x$

$\Rightarrow$ Radius of convergence is ∞.

2. What is the x^3 term in the Maclaurin series for $\sin x$?

$$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$$

$$= (-1)^0 \frac{x^1}{1} + (-1)\cdot\frac{x^3}{3!} + \dots$$

$$\Rightarrow \frac{-1}{3!} = \frac{-1}{3\times 2\times 1} = -\frac{1}{6}$$

3. The Taylor series of $f(x) = \frac{1}{1-x}$ at $x = 0$ is $\sum_{n=0}^{\infty} x^n$. $\Rightarrow$ Maclaurin series of $\frac{1}{1-x}$

What is the coefficient of x^3 in the Taylor series of $h(x) = \frac{1}{1-2x}$ at $x = 0$?

$$\frac{1}{1-x} = \sum_{n=1}^{\infty} x^n, \quad x = y \Rightarrow \frac{1}{1-y} = \sum_{n=1}^{\infty} y^n$$

$$y = 2x: \quad \frac{1}{1-2x} = \sum_{n=1}^{\infty} (2x)^n = (2x)^1 + (2x)^2 + \underline{(2x)^3} + \dots$$

$$8x^3 \Rightarrow 8$$

4. The MacLaurin series of $g(x) = \cos x$ is $\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n}$.

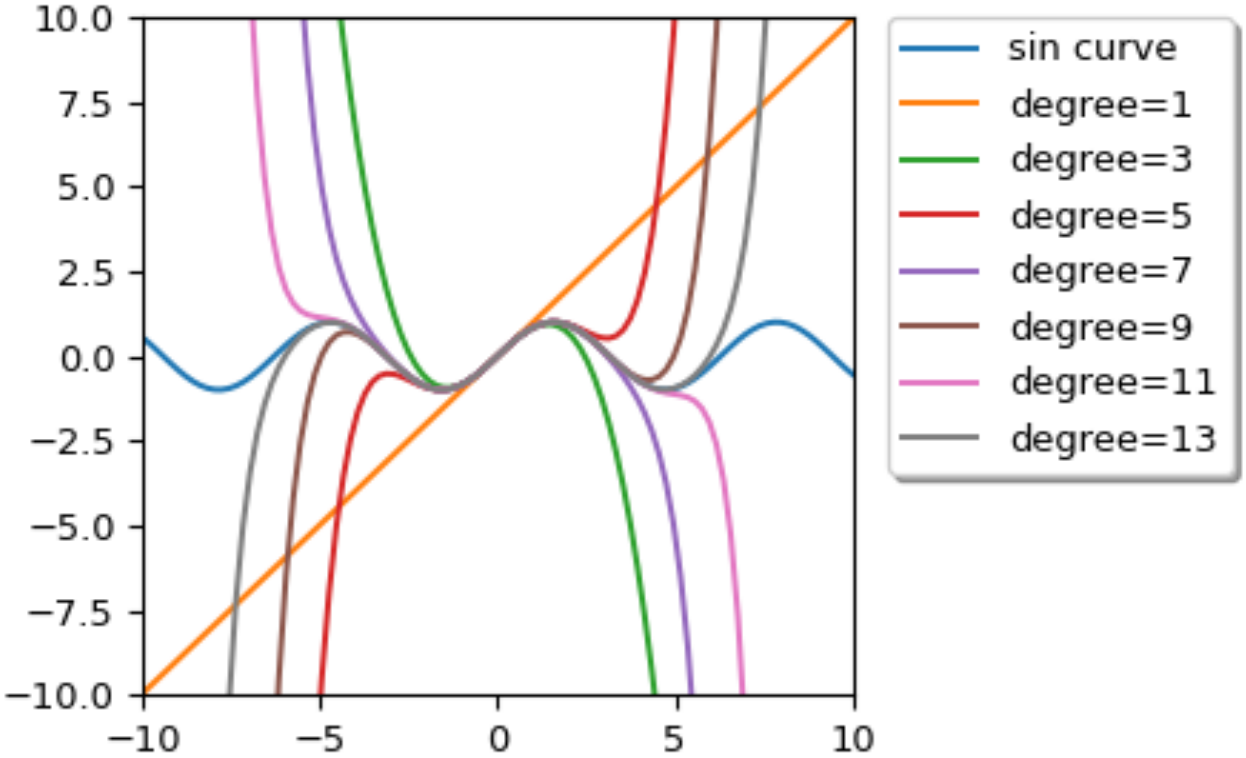
What is the Maclaurin series of $h(x) = x^2 \cos x$?

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} = 1 - \frac{x^2}{2} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

$$h(x) = x^2 \cdot \cos x = x^2 \cdot \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n}$$

$$= x^2\left(1 - \frac{x^2}{2} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots\right)$$

$$= x^2 - \frac{x^4}{2} + \frac{x^6}{4!} - \frac{x^{10}}{6!} + \dots$$


10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
−10
−5
0
5
10
sin curve
degree=1
degree=3
degree=5
degree=7
degree=9
degree=11
degree=13

Part 2: Problem Sheet 6, Power Series

Q1: Determine radius & interval of convergence for a given series. Discuss Q1a and 1d.

Q2: Taylor series from Taylor series formula; part (b) is a nice application of Taylor series.

Q3: New series from old (shortcuts !)

Q4: Taylor series in two ways (short way & long way). You are expected to know how to construct the Taylor series using the formula. Discuss this!

Q5: Determine higher derivatives of $f(x)$ if you already know the Taylor series of $f(x)$.

Q6: Past Exam question! An example of an exam question.

Problem Sheet 6, Q1 & 2

Q1 a, b Q2 b

15 mins

Question 1. Find the radius and interval of convergence for the following power series

a $\displaystyle\sum_{n=1}^{\infty} \frac{x^n}{\sqrt{n}}$ **(b)** $\displaystyle\sum_{n=0}^{\infty} \sqrt{n}\,(x-1)^n$ **(c)** $\displaystyle\sum_{n=2}^{\infty} \frac{x^n}{\ln n}$

d $\displaystyle\sum_{n=0}^{\infty} n!\,x^n$ **(e)** $\displaystyle\sum_{n=2}^{\infty} (-1)^n \frac{x^n}{4^n \ln n}$

Question 2.

(a) Find the terms up to and including the x^3 term in the Taylor series expansion of the function

$$f(x) = \ln\left(\frac{1}{1-x}\right)$$

about $x = 0$. Hence find a numerical approximation for $\ln 2$.

b, Find the first four non zero terms of the about $x = 0$ expansion of $f(x) = \sqrt{1+x}$. Hence, find a numerical approximation for $\sqrt{1.2}$. Hint: Binomial series

7.5 The p-test

For $p \in \mathbb{R}$, the p-series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is

convergent if $p > 1$
and divergent if $p \leq 1$.

If the alternating series

$$\sum_{n=0}^{\infty} (-1)^n b_n = b_0 - b_1 + b_2 - b_3 + \dots \quad (\text{all } b_n > 0)$$

satisfies

(i)	$\lim_{n\to\infty} b_n = 0$ and
(ii)	$b_{n+1} \leq b_n$, for all $n \geq 0$

then the series is convergent.

$$(1+x)^m = \sum_{n=0}^{\infty} \binom{m}{n} x^n. \qquad \binom{m}{n} = \frac{m!}{(m-n)!n!}.$$

Ratio test

Question 1. Find the radius and interval of convergence for the following power series

a $\sum_{n=1}^{\infty} \frac{x^n}{\sqrt{n}}$ **(b)** $\sum_{n=0}^{\infty} \sqrt{n}\,(x-1)^n$ **(c)** $\sum_{n=2}^{\infty} \frac{x^n}{\ln n}$

d $\sum_{n=0}^{\infty} n!\,x^n$ **(e)** $\sum_{n=2}^{\infty} (-1)^n \frac{x^n}{4^n \ln n}$

Question 2.

a. $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left|\frac{\cancel{x^{n+1}}}{\sqrt{n+1}} \cdot \frac{\sqrt{n}}{\cancel{x^n}}\right| = \lim_{n\to\infty} \left| x \cdot \sqrt{\frac{n}{n+1}} \right| = |x| \cdot \underbrace{\lim_{n\to\infty} \left|\sqrt{\frac{n}{n+1}}\right|}_{\to 1} = |x| \cdot 1 < 1$

$a_n = \frac{x^n}{\sqrt{n}}$

when $|x| < 1 \Rightarrow$ series is convergent ; when $|x| > 1 \Rightarrow$ divergent

$x = \pm 1$:

① $x = 1$: $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} \Rightarrow$ divergent by p-test , $p = \frac{1}{2}$

② $x = -1$: $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}} \Rightarrow$ convergent by the alternative test

Interval of convergence is $[-1, 1)$, radius is 1

d. $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left|\frac{\cancel{(n+1)!} \cdot \cancel{x^{n+1}}}{\cancel{n!} \cdot \cancel{x^n}}\right| = \lim_{n\to\infty} \left|(n+1)x\right| = |x| \cdot \lim_{n\to\infty} |n+1| = \infty$ for $x \neq 0 \Rightarrow$ divergent

$a_n = n! \cdot x^n$ If $x = 0$, $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = 0 < 1 \Rightarrow$ convergent

In conclusion, radius is 0 and convergent only when $x = 0$

b. Find the first four non zero terms of the about $x = 0$ expansion of $f(x) = \sqrt{1+x}$. Hence, find a numerical approximation for $\sqrt{1.2}$. Hint: $\boxed{\text{Binomial series}}$

$f(x) = (1+x)^m = \sum_{n=0}^{\infty} \binom{m}{n} x^n$, where $\binom{m}{n} = \frac{m!}{(m-n)! \cdot n!}$

$= 1 + mx + \frac{m(m-1)}{2} x^2 + \frac{m(m-1)(m-2)}{3!} x^3 + \dots$

$f(x) = \sqrt{1+x} = (1+x)^{\frac{1}{2}} = 1 + \frac{1}{2}x + \frac{\frac{1}{2}(\frac{1}{2}-1)}{2} x^2 + \frac{\frac{1}{2}(\frac{1}{2}-1)(\frac{1}{2}-2)}{3!} x^3 + \dots$

$= 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}x^3 + \dots$

$\sqrt{1.2} = \sqrt{1+0.2} = \sqrt{1+\frac{1}{5}}$

$x = \frac{1}{5}$

$f(\frac{1}{5}) = \sqrt{1+\frac{1}{5}} = (1+\frac{1}{5})^{\frac{1}{2}} \approx 1 + \frac{1}{2} \cdot \frac{1}{5} - \frac{1}{8} \cdot (\frac{1}{5})^2 + \frac{1}{16} \cdot (\frac{1}{5})^3 = \frac{2191}{2000} = 1.0955$

Part 3: Collaborative Exercise

- Max group size = 4
- Discuss the questions with other students at your table.
- Complete the collaborative exercise TOGETHER!
- Technology free assessment!
- Your tutors will check your answers and correct any errors. Feel free to move around and use the whiteboard.
- Submit the collaborative sheet and 'scrap' workings to your tutor.
- Remember to write your name, ID and AG (Allocated Applied Class Group #) on the hard copy.
- Note: It is academic misconduct to include names of students who are not present at the practical.

Marking Scheme for Collaborative Exercise

- Mark +0: Your group has no idea!
- Mark +1: Your group has some understanding but several areas require attention.
- Mark +2: Your group has a good understanding but some areas need attention.
- Mark +3: Your group has a very good understanding; minor issues
- Mark +4: Your group has demonstrated excellent understanding!

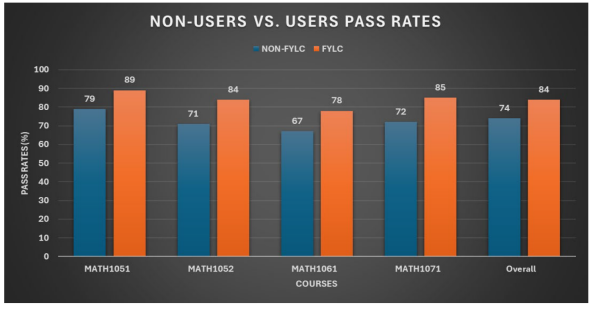
- Mark +1: Everyone in your group has contributed to completing the collaborative sheet. Workings on scrap paper will be collected by tutors in addition to the collaborative sheet.

- Maximum marks available is 5.

Remember…there is a lot of help available:

Help	Details
Support Learning Tutorial (from Week 2)	Wed 11-12, room 67-442.
First Year Learning Centre (FYLC)	Mon 11-2, Tue 1-4, Wed 1-4, Thu 11-2, Fri 1-4, 67-443.
Ed online discussion	24/7, staff response within 24 hours (except on weekends!)
Blackboard resources	24/7
Tutors	during applied classes

FYLC, SEMESTER 1, 2024



More info on `Blackboard > Course Information > MATH1051 Support`

Wrap up! Week 8 → Week 9

By the end of Week 8…

- Complete a selection of exercises from:
  - Competency Exercises – Power Series
  - Problem Sheet 6 – Power Series

Looking ahead to Week 9…

- Topic: Integration
- All resources for Week 9 are on Bb > Week 9 > Week 9 Applied Classes

Have a great week! See you in Week 9!